Spectral-Difference Low-Rank Representation Learning for Hyperspectral Anomaly Detection

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***Abstract*— Anomaly detection of a hyperspectral image without any prior information has attracted much more attention in remote sensing image understanding and interpretation, which aims at determining whether a sample belongs to background or anomaly. Low-rank dictionary learning plays an important role in exploiting the low-rank prior of background for hyperspectral image (HSI) anomaly detection. In this article, the low-rank dictionary learning is introduced to learn a dictionary which can reconstruct the background positively, while anomaly cannot. Considering the high correlation of data especially between the adjacent bands, we resort to spectral-difference low-rank dictionary representation learning for global background modeling which can fully exploit the low-rank prior of background. Then, the residual matrix is used to distinguish anomaly. Different from the existing anomaly detection methods based on dictionary which is constructed or learned in a separated step, our proposed model can simultaneously learn the dictionary and separate anomaly by iterative learning. The experimental results on five real data sets demonstrate the superior performance of the proposed method for hyperspectral anomaly detection compared with other state-of-the-art algorithms.**

***Index Terms*— Hyperspectral anomaly detection, low-rank dictionary learning, spectral-difference.**

I. Introduction

HYPERSPECTRAL image (HSI) is of high spectral resolution and abundant spatial information [1]–[4], which has superiority in distinguishing the ground object [5]–[7], and has been introduced to detection applications including hyperspectral target detection and anomaly detection [8]–[11]. In the case of high cost and difficulty in obtaining the supervision information, anomaly detection is popular with the field of precision agriculture, environmental monitoring, and military reconnaissance.

Hyperspectral anomaly detection aims at finding pixels or subpixels which have different spectral signatures compared with their neighbor background without any prior information. There are many challenges in hyperspectral anomaly detection, such as the lack of spectral prior about anomaly, complex background, and unbalanced distribution between the background and anomaly. At present, many researchers proposed various methods to tackle this problem, which include the statistic-based method, matrix-decomposition-based method, deep-learning-based method, and representation-based method. As the benchmark of the statistic-based method [12], RX proposed by Reed—Xiaoli assumes that the hyperspectral data obey a multivariate Gaussian distribution, and the mean value and the covariance of background from the whole image are estimated [13]. Different from global RX (GRX), the local RX (LRX) estimates the background by a local window around the test pixel. Then, the Mahalanobis distance between the test pixel and the background is computed to evaluate the anomalous level. To separate the background and anomaly, the matrix-decomposition-based methods have shown good prospect. Assuming that the background is low-rank and the anomalies are sparse, the low-rank and sparse decomposition model with the mixture of Gaussian model (LSDM-MoG) [14] was applied for HSI anomaly detection, in which the sparse component is modeled by the mixture Gaussian to describe the complex noise including anomaly, and a non-Gaussian detector based on the Manhattan distance is designed. However, it is inaccurate to model the background or anomaly using a specific distribution in reality.

With the advantage in capturing the high-level features, the methods based on deep learning have been explored for HSI anomaly detection. In [15], the autoencoder (AE) network was applied to reconstruct the background, and the error between the original image and the decoder image was used to measure the anomaly. Lu *et al.* [16] introduced the manifold constraint in the AE network, which can exploit the local information in background reconstruction to detect anomalies. In [17], the selected pixels via unsupervised background searching were used to train the AE with the discriminator network to learn the background distribution. Li *et al.* [10] introduced the transfer learning to train the convolutional neural network for HSI anomaly detection.

Recently, the methods based on representation have been successfully applied in HSI analysis [18]–[21]. Sparse representation is to find the sparsest linear combinations of dictionary atoms for each vector of data individually [22], [23]. In the case of the existing spectral prior, Zhu *et al.* proposed to construct the target dictionary based on constrained energy minimization (CEM) [24] and the background

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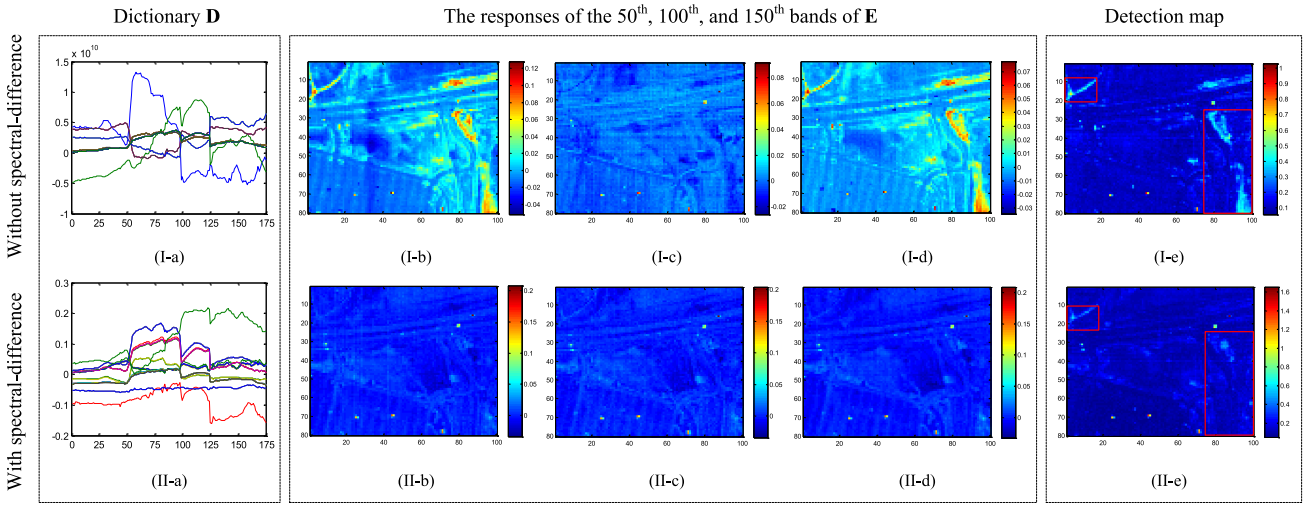


Fig. 1. Comparisons between low-rank regularization and spectral-difference low-rank regularization. The first row illustrates the low-rank regularization case, and the second row shows the spectral-difference low-rank regularization case. (a) Characteristic curves of the learned dictionary. The magnitudes of the dictionary atoms in (II-a) are more stable than in (I-a). (b)–(d) Responses of the 50th, 100th, and 150th bands of **E**. The complex background can be better compressed from anomaly in (II-b)–(II-d) than in (I-b)–(I-d). (e) Detection map. For the areas of the red tangles, the background is better compressed in (II-e) than in (I-e).

dictionary based on the dual concentric window to formulate the sparse-representation-based target detection model [25]. An HSI anomaly detection method based on sparse representation linear mixture model (SR-LMM) [26] is proposed with the assumption that the background can be sparsely represented by its spatial neighbors in the absence of spectral prior. Since the obtained data in HSI are redundant and highly correlated, the low-rank representation (LRR) [27] is usually exploited to capture the global information of data. The LRR model assumes that the data are distributed in multiple underlying subspaces, and each vector of data can be regarded as the lowest rank linear representation with the same dictionary. A new abundance and dictionary-based low-rank representation (ADLR) method [28] is proposed for HSI anomaly detection. Considering that the local geometrical information and spatial information are of great importance for the discrimination of background and anomaly, Cheng and Wang [29] introduced the graph regularization and total variation (TV) regularization into the LRR model for HSI anomaly detection. Xu *et al.* [30] proposed the low-rank and sparse representation (LRASR) method for the HSI anomaly detection, in which a clustering-based dictionary construction algorithm is presented to describe the background, and the representation coefficients matrix is obtained by low-rank regularization and sparse regularization to explore the global and local structure information. To better separate the background and anomaly, Huyan *et al.* [31] introduced the potential strategy to construct anomaly dictionary for low-rank and sparse representation model. In these methods, the sparse regularization can be used to represent the local information and the low-rank regularization can capture the global structure information.

In the representation-based methods, dictionary plays an important role for data reconstruction. As the complexity of background which contains multiple land covers with different spectra, the methods of dictionary construction are difficult to ensure that the dictionary can comprehensively reflect the complex spectral features about background. Therefore, dictionary learning captures our attention [32], [33]. Supposing that the pixels in neighborhood are more likely to belong to the same class and share the same sparse pattern, the patch-based contextual dictionary learning method [34] was proposed to obtain the discriminative sparse coefficients for HSI classification. With the development of low-rank recovery theory, the low-rank representation-based dictionary learning methods have been explored [35], [36]. In hyperspectral target detection [37], the low-rank constraint and the incoherent constraint with the target spectra were applied to the background dictionary learning. The existing methods based on low-rank dictionary learning (LRDL) have been explored for supervised learning of HSI.

Inspired by the assumption that the background usually follows a low-rank property and anomalies are distributed sparsely, in this article, the LRDL is introduced to learn a dictionary which can be positively used for background reconstruction. The anomaly can be demonstrated from the residual part. In the formulation, the background is factorized into a dictionary and a coefficients matrix. The sparse regularization is adopted on the representation coefficients matrix so that the pixels can be linearly represented by the learned dictionary using a few atoms. As the background is the dominant component of the whole image, the dictionary is obtained by learning and optimized from the whole image rather than several selected pixels, and the learned dictionary can reflect the global characteristic of the data. To exploit the low-rank characteristic of background, the low-rank constraint is applied to the dictionary. Fig. 1(I-a) shows the characteristic curves of the learned dictionary via LRDL, but the magnitude of the curves changes dramatically. As the HSI contains hundreds of bands, there is serious redundancy,

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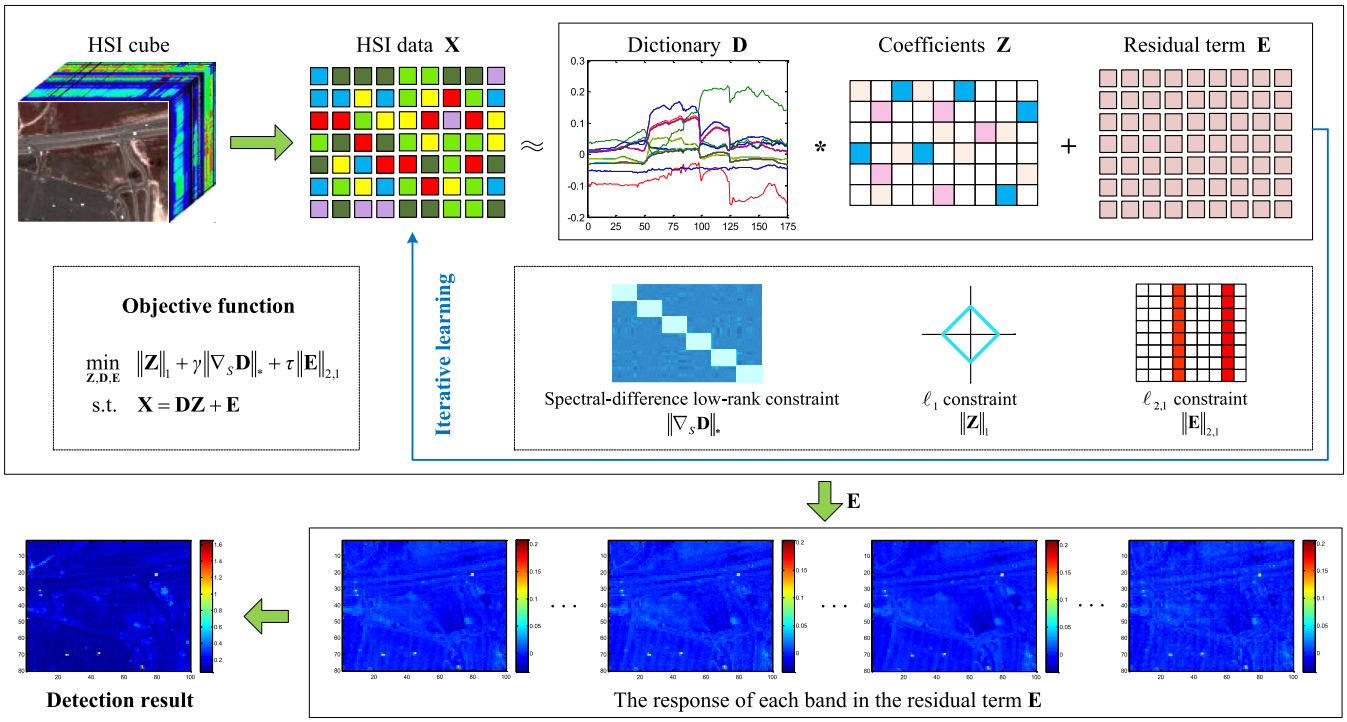


Fig. 2. Illustration of the proposed spectral-difference low-rank dictionary representation learning for HSI anomaly detection.

especially on the adjacent bands. To further explore the low-rank characteristic of spectral contextual information, the spectral-difference LRDL (SDLRDL) is introduced in this article. Fig. 1 demonstrates the characteristics with and without spectral-difference regularization. The whole scheme of the proposed spectral-difference representation learning method for HSI anomaly detection is illustrated in Fig. 2.

The contributions of our proposed hyperspectral anomaly detection method based on spectral-difference low-rank representation learning can be summarized as follows.

1) We introduce the LRDL for HSI anomaly detection, in which the low-rank regularization is applied on the dictionary to capture the global structure for the reconstruction of the background and detect anomaly in hyperspectral image.
2) To enhance the contextual low-rank constraint between the adjacent bands, the spectral-difference low-rank dictionary leaning model is proposed for better anomaly detection.
3) The experimental results on five real HSI data sets illustrate that the proposed SDLRDL method outperforms other state-of-the-art algorithms in hyperspectral anomaly detection.

The remainder of this article is organized as follows. Section II gives the basic knowledge about sparse representation and low-rank representation. Section III provides the detailed description of the proposed spectral-difference low-rank dictionary representation learning method for HSI anomaly detection. Section IV presents the experimental results and analysis, and the conclusions and the future research directions are given in Section V.

II. Related Work

A. Sparse Representation

Sparse representation has been widely studied in image processing [38]. For a given dictionary, the signals are the sparsest linearly represented via atoms of the dictionary with sparsity regularization. The objective function can be explained by solving the following optimization problem:

$$\min_{\mathbf{z}} \|\mathbf{z}\|_0 \quad \mathbf{s.t.}\ \mathbf{x} = \mathbf{Dz} \tag{1}$$

where $\|\cdot\|_0$ denotes the ℓ_0 norm, which is used to evaluate the number of nonzero elements of a matrix. $\mathbf{D}$ is a dictionary, $\mathbf{z}$ is the sparse coefficients vector, and $\mathbf{x}$ is a test signal. As the problem of (1) is an NP-hard problem, it can be relaxed as the following form:

$$\min_{\mathbf{z}} \|\mathbf{z}\|_1 \quad \mathbf{s.t.}\ \mathbf{x} = \mathbf{Dz} \tag{2}$$

where $\|\cdot\|_1$ denotes the ℓ_1 norm to compute the sum of the absolute value of all the nonzero entries in the matrix.

B. Low-Rank Representation

The low-rank model supposes that a matrix $\mathbf{X}$ can be decomposed into two matrices, that is, $\mathbf{X} = \mathbf{A} + \mathbf{E}$, where $\mathbf{A}$ is a low-rank matrix and $\mathbf{E}$ is a noise matrix [39]. In LRR, Liu *et al.* [27] introduced data itself as the dictionary to better segment the data into their respective subspaces, so the self-representation low-rank model can be expressed as

$$\min_{\mathbf{Z},\mathbf{E}} \|\mathbf{Z}\|_* + \lambda\|\mathbf{E}\|_{2,1} \quad \text{s.t. } \mathbf{X} = \mathbf{XZ} + \mathbf{E} \tag{3}$$

where $\|\cdot\|_*$ denotes the nuclear norm of a matrix, that is, the sum of the singular values, $\|\mathbf{Z}\|_* = \sum_i \sigma_i(\mathbf{Z})$, where σ_i

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denotes the singular value of $\mathbf{Z}$. $\|\cdot\|_{2,1}$ is called as the $\ell_{2,1}$ norm. λ is a parameter to balance the low-rank term and the $\ell_{2,1}$ norm term.

III. Spectral-Difference Low-Rank Representation Learning for Hyperspectral Anomaly Detection

Generally, the HSI cubic can be converted as a matrix and denoted as $\mathbf{X}$. To distinguish the background and anomaly of the whole image, the dictionary-based representation method is adopted, in which the product of a dictionary and the corresponding coefficients matrix are used to describe the background. So the model can be formulated as $\mathbf{X} = \mathbf{DZ} + \mathbf{E}$, where $\mathbf{DZ}$ represents the background, $\mathbf{D}$ is the dictionary, $\mathbf{Z}$ is the coefficients matrix, and $\mathbf{E}$ is the residual part.

A. Low-Rank Dictionary Learning

Traditional background dictionary is constructed by selecting pixels from the HSI data, which makes sure that the spectra of different ground objects can be included. Since the spectra of homogeneous ground objects would be different and the heterogeneous materials may have the same spectrum, the background is complex and the constructed dictionary cannot ensure the selected atoms are enough to reflect the background spectral characteristics comprehensively. To capture the global structure information about background, we formulate the estimation of background as the product of the dictionary and the coefficients matrix and introduce the low-rank regularization into dictionary learning. Meanwhile, instead of adopting a fixed dictionary which is constructed or learned in a separated step, we update the dictionary and coefficients alternatively to learn a more comprehensive dictionary for background reconstruction.

The low-rank dictionary representation learning model can be formulated as

$$\begin{aligned} &\min_{\mathbf{Z},\mathbf{D},\mathbf{E}} \ \|\mathbf{Z}\|_1 + \alpha\|\mathbf{D}\|_* + \beta\|\mathbf{E}\|_{2,1} \\ &\text{s.t. } \mathbf{X} = \mathbf{DZ} + \mathbf{E} \end{aligned} \tag{4}$$

where $\mathbf{X} \in R^{B\times n}$, B is the number of bands, and n is the number of pixels. $\mathbf{D} \in R^{B\times m}$ is the dictionary, which includes m atoms. In general, background is the dominant component and redundant. The data of background pixels are approximately distributed in a low-rank space, and the anomalies are sparse. We hope the learned dictionary $\mathbf{D}$ is with discriminating and reconstructive capabilities, which can reconstruct the background positively, while the anomaly cannot. The nuclear norm constraint on the dictionary can well-explore the low-rank prior of data for background reconstruction. Thus, the low-rank constraint is applied on $\mathbf{D}$ in (4). The dictionary is learned from the whole data, and it can reflect the global characteristic of the data. $\mathbf{Z} \in R^{m\times n}$ is the coding coefficients matrix. With the sparse regularization, the data can be linearly represented by the dictionary atoms, and the ℓ_1 norm is applied on $\mathbf{Z}$. $\mathbf{E} \in R^{B\times n}$ is the residual part, which includes Gaussian noise and the reconstruction error about background and anomaly. Through LRDL, the reconstruction error of background is far less than anomaly, and the anomalies can be demonstrated from the discriminating matrix $\mathbf{E}$. Fig. 3 illustrates three types of errors [40]. For the sparse noise in Fig. 3(b), the ℓ_1 norm is applicable. While in the representation based on 2-D matrix, the anomalies are sample-specific which are more similar to Fig. 3(c). The $\ell_{2,1}$ norm can explore the structural sparsity and encourages the columns of matrix to be zero, and thus the $\ell_{2,1}$ norm is applied on matrix $\mathbf{E}$. α and β are the two positive weighting parameters. In (4), $\|\cdot\|_{2,1}$ is the $\ell_{2,1}$ norm and it can be computed by

$$\|\mathbf{E}\|_{2,1} = \sum_{j=1}^{n} \sqrt{\sum_{i=1}^{B} ([\mathbf{E}]_{i,j})^2} \tag{5}$$

where $[\mathbf{E}]_{i,j}$ is the entry of $\mathbf{E}$.

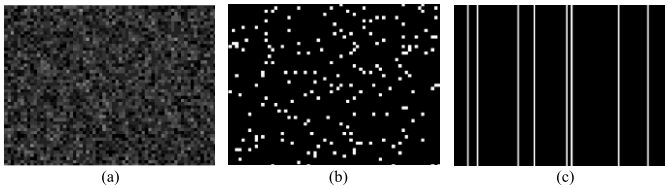


Fig. 3. (a) Gaussian noise. (b) Random corruptions. (c) Sample-specific corruptions.

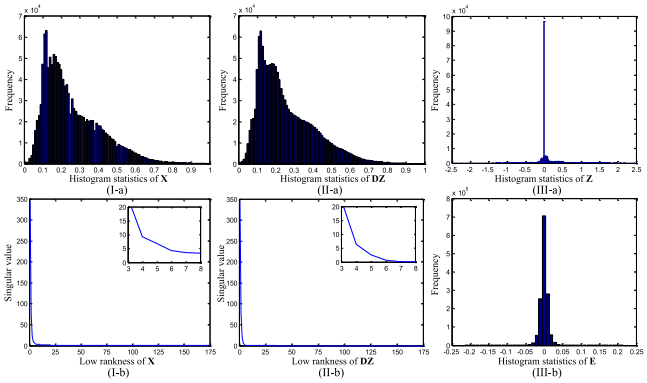


Fig. 4. (I-a) and (II-a) Histogram statistics of $\mathbf{X}$ and $\mathbf{DZ}$ (in HYDICE Urban), respectively, and we can observe that the SDLRDL model can reconstruct the main component of the original data. (I-b) and (II-b) Low rankness of $\mathbf{X}$ and $\mathbf{DZ}$, respectively. The low rankness in spectral-different space is reinforced which can be observed via comparing (II-a) and (II-b). (III-a) Histogram statistics of $\mathbf{Z}$, which demonstrates that the coefficients matrix $\mathbf{Z}$ is sparse as most of the coefficients in $\mathbf{Z}$ are zero. (III-b) Histogram statistics of $\mathbf{E}$, which demonstrates that most pixels can be well-reconstructed with small reconstruction errors.

B. Spectral-Difference Low-Rank Dictionary Learning

In the dictionary-learning-based representation model of (4), the spectral features of background are included in the dictionary. In the original data space of HSI, there are redundancy, especially on the adjacent bands. To explore the contextual information of spectral, the SDLRDL method is proposed. This model can be represented as follows:

$$\begin{aligned} &\min_{\mathbf{Z},\mathbf{D},\mathbf{E}} \ \|\mathbf{Z}\|_1 + \gamma\,\|\nabla_S\mathbf{D}\|_* + \tau\|\mathbf{E}\|_{2,1} \\ &\text{s.t. } \mathbf{X} = \mathbf{DZ} + \mathbf{E} \end{aligned} \tag{6}$$

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where γ and τ are the two weighting parameters, $\nabla_S\mathbf{D}$ denotes the spectral-difference operator of $\mathbf{D}$, and the definition is

$$\nabla_S\mathbf{D}(i,:)=\begin{cases}0, & i=1\\ \mathbf{D}(i,:)-\mathbf{D}(i-1,:), & i=2,3,\ldots,B.\end{cases}\tag{7}$$

In (7), the difference between the adjacent bands is computed. The dictionary $\mathbf{D}$ with spectral-difference low-rank constraint considered the redundancy between the adjacent bands and the low rankness of background. $\mathbf{D}$ is learned from the whole data and optimized in global, and thus the dictionary can learn the global low-rank features to reconstruct background. The histogram statistics analysis about the proposed model is presented in Fig. 4. In a histogram, the y-axis is the number of data points that lie within a range of values as grouped in the x-axis. The learned dictionary $\mathbf{D}$ can reflect the low-rank features of the data. Compared with the LRDL in (4), the dictionary learned by SDLRDL of (6) is more stable, and the obtained residual matrix $\mathbf{E}$ is more discriminative in anomaly detection, as shown in Fig. 1.

C. Optimization and Complexity Analysis

To solve the optimization problem of (6), we introduce an auxiliary variable $\mathbf{H}$ to make the variables separable. The problem of (6) can be rewritten as

$$\begin{aligned}&\min_{\mathbf{Z},\mathbf{D},\mathbf{E}}\ \|\mathbf{Z}\|_1+\gamma\|\mathbf{H}\|_*+\tau\|\mathbf{E}\|_{2,1}\\ &\text{s.t. }\mathbf{X}=\mathbf{DZ}+\mathbf{E},\nabla_S\mathbf{D}=\mathbf{H}.\end{aligned}\tag{8}$$

The augmented Lagrangian function of (8) is

$$\begin{aligned}&L(\mathbf{Z},\mathbf{D},\mathbf{E},\mathbf{H},\mathbf{T}_1,\mathbf{T}_2,\mu)\\ &=\|\mathbf{Z}\|_1+\gamma\|\mathbf{H}\|_*+\tau\|\mathbf{E}\|_{2,1}\\ &\quad+\langle\mathbf{T}_1,\mathbf{X}-\mathbf{DZ}-\mathbf{E}\rangle+\langle\mathbf{T}_2,\nabla_S\mathbf{D}-\mathbf{H}\rangle\\ &\quad+\frac{\mu}{2}\left(\|\mathbf{X}-\mathbf{DZ}-\mathbf{E}\|_F^2+\|\nabla_S\mathbf{D}-\mathbf{H}\|_F^2\right)\\ &=\|\mathbf{Z}\|_1+\gamma\|\mathbf{H}\|_*+\tau\|\mathbf{E}\|_{2,1}\\ &\quad+\frac{\mu}{2}\left(\|\mathbf{X}-\mathbf{DZ}-\mathbf{E}+\mathbf{T}_1/\mu\|_F^2+\|\nabla_S\mathbf{D}-\mathbf{H}+\mathbf{T}_2/\mu\|_F^2\right)\\ &\quad-\frac{1}{2\mu}\left(\|\mathbf{T}_1\|_F^2+\|\mathbf{T}_2\|_F^2\right)\end{aligned}\tag{9}$$

where $\mathbf{T}_1$ and $\mathbf{T}_2$ are the Lagrange multipliers, and $\mu>0$ is a penalty parameter. The variables are alternately updated by minimizing the augmented Lagrangian function with other variables fixed. We provide the details of solving (9) with alternating direction method of multipliers (ADMM) in the following.

1) Fix $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{H}$ and update $\mathbf{Z}$ as follows:

$$\begin{aligned}\mathbf{Z}&=\arg\min_{\mathbf{Z}}\|\mathbf{Z}\|_1+\langle\nabla_{\mathbf{Z}}f(\mathbf{D},\mathbf{Z},\mathbf{E},\mathbf{T}_1,\mu),\mathbf{Z}-\mathbf{Z}_k\rangle\\ &\quad+\frac{\eta_1\mu}{2}\|\mathbf{Z}-\mathbf{Z}_k\|_F^2\\ &=\arg\min_{\mathbf{Z}}\|\mathbf{Z}\|_1+\frac{\eta_1\mu}{2}\|\mathbf{Z}-\mathbf{Z}_k-\mathbf{D}^T(\mathbf{X}-\mathbf{DZ}-\mathbf{E}\\ &\qquad+\mathbf{T}_1/\eta_1\|_F^2\end{aligned}\tag{10}$$

where $f(\mathbf{D},\mathbf{Z},\mathbf{E},\mathbf{T}_1,\mu)=\frac{\mu}{2}\|\mathbf{X}-\mathbf{DZ}-\mathbf{E}+\mathbf{T}_1/\mu\|_F^2$, and it is replaced by the first-order approximation at the previous iteration and adding a proximal term [41]. $\nabla_{\mathbf{Z}}f$ is the partial differential of f respecting $\mathbf{Z}$, $\eta_1=\|\mathbf{D}\|_2^2$.

2) Fix $\mathbf{Z}$, $\mathbf{E}$, and $\mathbf{H}$ and update $\mathbf{D}$ as follows:

$$\begin{aligned}\mathbf{D}&=\arg\min_{\mathbf{D}}\langle\nabla_{\mathbf{D}}f(\mathbf{D},\mathbf{Z},\mathbf{E},\mathbf{T}_1,\mu),\mathbf{D}-\mathbf{D}_k\rangle\\ &\quad+\frac{\eta_2\mu}{2}\|\mathbf{D}-\mathbf{D}_k\|_F^2+\frac{\mu}{2}\|\nabla_S\mathbf{D}-\mathbf{H}+\mathbf{T}_2/\mu\|_F^2\\ &=\arg\min_{\mathbf{D}}\frac{\eta_2\mu}{2}\|\mathbf{D}-\mathbf{D}_k+[-(\mathbf{X}-\mathbf{DZ}-\mathbf{E}+\mathbf{T}_1/\mu)\mathbf{Z}^T]\\ &\quad/\eta_2\|_F^2+\frac{\mu}{2}\|\nabla_S\mathbf{D}-\mathbf{H}+\mathbf{T}_2/\mu\|_F^2\end{aligned}\tag{11}$$

where $\nabla_{\mathbf{D}}f$ is the partial differential of f respecting $\mathbf{D}$, $\eta_2=\|\mathbf{Z}\|_2^2$.

3) Fix $\mathbf{Z}$, $\mathbf{D}$, and $\mathbf{H}$ and update $\mathbf{E}$ as follows:

$$\mathbf{E}=\arg\min_{\mathbf{E}}\tau\|\mathbf{E}\|_{2,1}+\frac{\mu}{2}\|\mathbf{X}-\mathbf{DZ}-\mathbf{E}+\mathbf{T}_1/\mu\|_F^2.\tag{12}$$

4) Fix $\mathbf{Z}$, $\mathbf{D}$, and $\mathbf{E}$ and update $\mathbf{H}$ as follows:

$$\mathbf{H}=\arg\min_{\mathbf{H}}\gamma\|\mathbf{H}\|_*+\frac{\mu}{2}\|\nabla_S\mathbf{D}-\mathbf{H}+\mathbf{T}_2/\mu\|_F^2.\tag{13}$$

In (11), as the difference operator $\nabla_S\mathbf{D}$ is the convolution computation of $\mathbf{D}$ with $[1,-1]^T$, we introduce Fourier transform to update $\mathbf{D}$ as (14), as shown at the bottom of the next page, where $\mathcal{F}$ and $\mathcal{F}^{-1}$ are the Fourier transform and Fourier inverse operations, respectively. $\mathcal{F}(\cdot)^H$ is the conjugate function.

The SDLRDL algorithm is summarized in Algorithm 1, where Ω, Ψ, and Θ are the singular value thresholding [42], $\ell_{2,1}$ minimization operator [27], and shrinkage operator [43] respectively.

Algorithm 1 Spectral-Difference Low-Rank Dictionary Learning Algorithm

Input: HSI data set $\mathbf{X}\in R^{B\times n}$, Dictionary atoms number m
Initialize: $\mathbf{Z}=\mathbf{D}=\mathbf{I}$, $\mathbf{E}=0$, $\mathbf{H}=\mathbf{T}_{1,0}=\mathbf{T}_{2,0}=0$, $\mu_{\max}=10^6$, $\mu=10^{-6}$, $\rho=1.1$, $\varepsilon=10^{-6}$ **While** not converged do

1. Fix others and update $\mathbf{Z}$ by
   $\mathbf{Z}:=\Theta_{(\eta_1\mu)^{-1}}(\mathbf{Z}_k+\mathbf{D}^T(\mathbf{X}-\mathbf{DZ}-\mathbf{E}+\mathbf{T}_1/\mu)/\eta_1)$
2. Fix others and update $\mathbf{D}$ as (14)
3. Fix others and update $\mathbf{E}$ by
   $\mathbf{E}:=\Psi_{\tau\mu^{-1}}(\mathbf{X}-\mathbf{DZ}+\mathbf{T}_1/\mu)$
4. Fix others and update $\mathbf{H}$ by
   $\mathbf{H}:=\Omega_{\gamma\mu^{-1}}(\nabla_S\mathbf{D}+\mathbf{T}_2/\mu)$
5. Update $\mathbf{T}_1$ and $\mathbf{T}_2$ by
   $\mathbf{T}_1:=\mathbf{T}_1+\mu(\mathbf{X}-\mathbf{DZ}-\mathbf{E})$
   $\mathbf{T}_2:=\mathbf{T}_2+\mu(\nabla_S\mathbf{D}-\mathbf{H})$
6. Update μ by
   $\mu=\min\{\rho\mu,\mu_{\max}\}$
7. Check the convergence conditions
   $\|\mathbf{X}-\mathbf{DZ}-\mathbf{E}\|_F<\varepsilon$
   $\|\nabla_S\mathbf{D}-\mathbf{H}\|_F<\varepsilon$

End while
Output: $\mathbf{Z}$, $\mathbf{D}$, $\mathbf{E}$, $\mathbf{H}$.

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Then, the response of the anomaly for each pixel is calculated as follows:

$$r(\mathbf{x}_j) = \left\|\mathbf{E}_{:,j}\right\|_2 = \sqrt{\sum_i ([\mathbf{E}]_{i,j})^2}, \quad j = 1, 2, \ldots, n. \tag{15}$$

In Algorithm 1, the computation complexity of solving $\mathbf{Z}$ is about $\mathcal{O}(m^2Bn)$ via the shrinkage operator. The major computational cost in solving $\mathbf{D}$ is the matrix multiplication and the complexity of solving $\mathbf{D}$ is about $\mathcal{O}(mBn)$. The complexity of solving $\mathbf{E}$ by the $\ell_{2,1}$ minimization operator is about $\mathcal{O}(B^2n)$. We use the singular value decomposition (SVD) to update $\mathbf{H}$and the complexity of solving $\mathbf{H}$ is about $\mathcal{O}(m^2B)$. Therefore, the total computation complexity of Algorithm 1 is about $\mathcal{O}\omega(m^2+m+B)Bn$, where ω is the number of iterations.

IV. Experimental Results

A. Data Sets

In the experiments, we apply our methods on five hyperspectral data sets to verify the performances of low-rank dictionary representation learning-based models. The details of the five data sets are described as follows.

HYDICE Urban Data Set [44]: It was captured by the hyperspectral digital imagery collection experiment (HYDICE) airborne sensor from an aircraft platform with a spectral resolution of 10 nm and a spatial resolution of 1 m approximately. The spatial size used in the experiments is 80 $\times$ 100 with 175 spectral bands which have removed the low SNR and vapor absorption bands. There are 21 anomalous pixels in the HYDICE Urban data set.

AVIRIS San Diego Data Set [30]: This data set was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor from San Diego with a spatial resolution of 3.5 m and the wavelengths ranging from 370 to 2510 nm. The spatial size used in the experiments is 100 $\times$ 100 with 186 spectral bands where the low SNR, water absorption regions, and bad bands have been removed.

Cri hyperspectral Data Set [45]: It was collected by the Nuance Cri hyperspectral sensor with a spectral resolution of 10 nm and the wavelengths ranging from 650 to 1100 nm. The spatial size used in the experiments is 400 $\times$ 400 with 46 spectral bands. There are 2216 anomalous pixels in the Cri hyperspectral data set.

MUUFL Gulfport Data Set [46], [47]: It was collected by the CASI-1500 sensor from an aircraft platform over the University of Southern Mississippi Gulf Park Campus in November 2010 with the wavelengths ranging from 375 to 1050 nm. The spatial size selected for the experiment is 325 $\times$ 220 with 64 bands. There exist 269 anomalous pixels to be detected in the MUUFL Gulfport data set.

Mosaic Avon Data Set [48]: It was collected by the airborne sensor of ProSpecTIR sensor system over a driving park in Avon, south of Rochester, NY, USA. The spatial resolution of this data set is 1 m and it covers the wavelengths ranging from 400 to 2450 nm. The spatial size selected for the experiment is 350 $\times$ 330 with 360 bands. There are 80 anomalous pixels to be detected including three red or blue felts, 25 tarps arranged as a grid, and ten people.

The pseudo-color images and the reference maps of the five data sets are shown in Fig. 5.

B. Detection Performance

In the experiment, six state-of-the-art methods are selected for comparison. As the benchmark of the statistic methods, global RX and local RX methods [13] are included. The LSDM-MoG method [14] which is based on the low-rank and sparse decomposition is adopted, and the deep learning method based on AE is selected [15]. In addition, the methods based on low-rank representation such as ADLR [28] and LRASR [30] are also chosen for evaluation. The receiver operating characteristic (ROC) curves are shown to reflect the relations between the probability of detection and false alarm rate, and the values of the area under curve (AUC) are presented to visually compare the detection results about different algorithms [49]. In addition, the detection result maps are shown to describe the response of background and anomaly.

First, the AUC values obtained by different methods on the five data sets are presented in Table I. For the method based on the statistic of GRX, the AUC values are all more than 0.90 on different data sets which are benefited by the estimation of background statistical characteristics from global. The LRX method performs better than the others on the first two data sets, but the performance of LRX is sensitive to the window sizes which are used to estimate the local background. For the LSDM-MoG method, the AUC values are pleasurable especially on the first two data sets which are both more than 0.95, as applying a mixture Gaussian to model the sparse component can better distinguish anomaly and noise. The AE-based method obtains positive results which exploits the nonlinear features of the data. The ADLR method demonstrates the acceptable detection results, which exploits the low-rank characteristic of background. The LRASR method obtains a good detection result on AVIRIS San Diego data set in that the constructed dictionary can better depict the background characteristic to detect anomalies. As the area of anomaly in Cri data set is relatively big, the clustering-based dictionary construction strategy in LRASR could not ensure that the background dictionary is without anomalous pixels, and this will decrease the detection value. For the LRDL method, the AUC values about the five data sets are all more than 0.94, and our dictionary learning-based methods demonstrate superior results. In our models, LRDL is the basic LRDL model. The SDLRDL method can fully exploit the low-rank characteristic of background and obtains the best

$$\mathbf{D} = \mathcal{F}^{-1}\left(\frac{\mathcal{F}(\eta_2(\mathbf{D}_k + (\mathbf{X} - \mathbf{DZ} - \mathbf{E} + \mathbf{T}_1/\mu)\mathbf{Z}^T/\eta_2)) + \mathcal{F}(\nabla_S)^H\mathcal{F}(\mathbf{H} - \mathbf{T}_2/\mu)}{\eta_2 + \mathcal{F}(\nabla_S)^T\mathcal{F}(\nabla_S)}\right) \tag{14}$$

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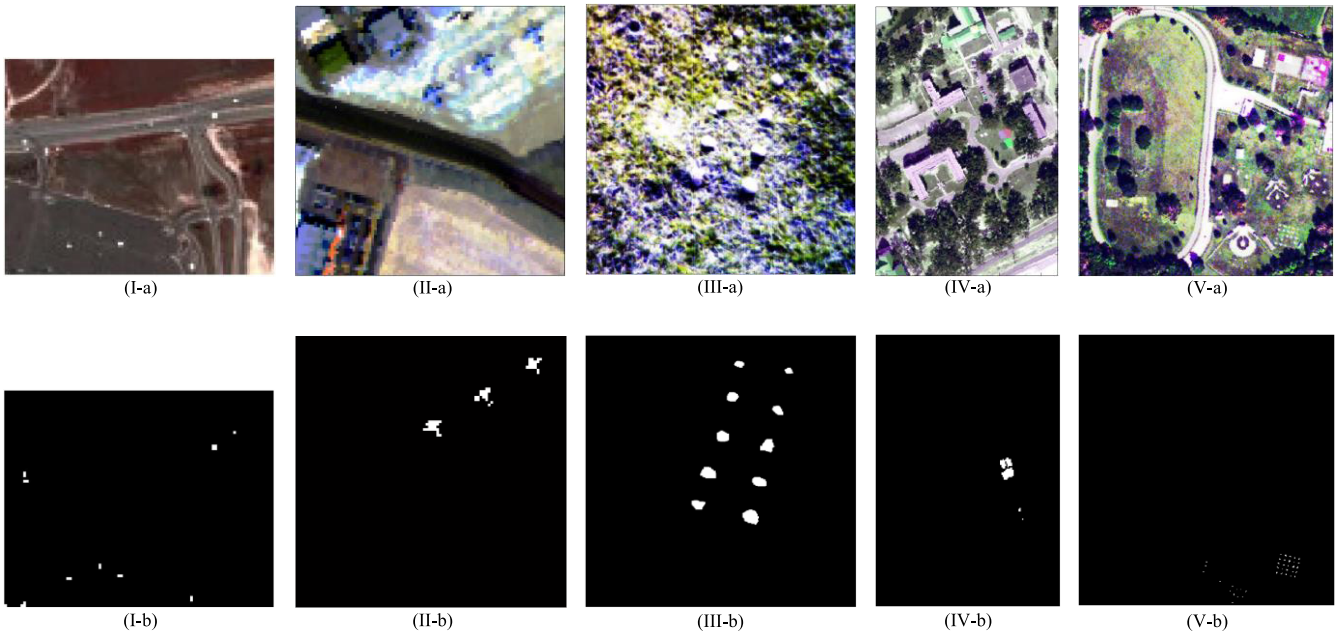


Fig. 5. Pseudo-color images and reference maps of the five hyperspectral data sets. (I-a) and (I-b) Pseudo-color image and the reference of the HYDICE Urban data set. (II-a) and (II-b) Pseudo-color image and the reference of the AVIRIS San Diego data set. (III-a) and (III-b) Pseudo-color image and the reference of the Cri hyperspectral data set. (IV-a) and (IV-b) Pseudo-color image and the reference of the MUUFL Gulfport data set. (V-a) and (V-b) Pseudo-color image and the reference of the Mosaic Avon data set.

TABLE I
AUC VALUES OBTAINED BY DIFFERENT METHODS ON THE FIVE DATA SETS

	HYDICE Urban	AVIRIS San Diego	Cri	MUUFL Gulfport	Mosaic Avon
GRX	0.9857	0.9090	0.9135	0.9321	0.9431
LRX	0.9492	0.9434	0.8290	0.8420	0.8173
LSDM-MoG	0.9676	0.9512	0.8305	0.9562	0.9199
AE	0.9306	0.9788	0.9373	0.9766	0.6937
ADLR	0.9255	0.9024	0.8628	0.9441	0.7624
LRASR	0.9489	0.9882	0.8298	0.9859	0.6866
LRDL	0.9760	0.9751	0.9799	0.9965	0.9406
SDLRDL	**0.9921**	**0.9885**	**0.9887**	**0.9985**	**0.9670**

result compared with other methods, as shown in Table I. On the first four data sets, our methods obtained desirable results, especially on Cri data set, and our proposed methods including LRDL and SDLRDL are obviously superior to the other methods as our methods can fully exploit the prior information such as low-rank of background and sparsity of anomalies. For Mosaic Avon data set, the performance of our proposed methods is acceptable in detecting small anomaly.

To further distinguish the anomaly visually, the detection maps about the five data sets are shown in Fig. 6. Fig. 6(a)–(h) are the detection maps about different methods, where the area with deeper color demonstrates background and the pixels with brighter color demonstrate anomalies. For GRX and LRX methods on the five data sets, the anomalies cannot be completely detected in the detection maps, which demonstrates the limitations of the distribution hypothesis for real data sets. For the LSDM-MoG method on different data sets from Fig. 6(I-c)–(V-c), the contrast between the background and the anomaly is not evident as the background is not well-compressed by this model. The AE-based method can reconstruct the background via nonlinear representation to detect anomaly, and the detection result verifies its effectiveness especially in Fig. 6(II-d)–(V-d). For the ADLR method, the detection maps are not satisfactory. Especially in Fig. 6(III-e)–(V-e), the visualization of the background regions are complex, since the detection results rely on unmixing and the background dictionary cannot get rid of the interference of anomaly. The LRASR method shows credible detection performance on the five data sets in that the clustering-based dictionary construction strategy plays a positive role. Fig. 6(g) and (h) are the detection maps of our proposed methods based on dictionary learning, in which we can distinguish the anomalies from background unambiguously. For the SDLRDL method of Fig. 6(I-h)–(V-h), the background area is with deeper color and the anomalies are with brighter color, which is benefited by the spectral-difference low-rank regularization that can better compress the background component to detect anomalies.

Moreover, the ROC curves of different methods on the five data sets are shown in Fig. 7. From the ROC curves on the HYDICE Urban data set of Fig. 7(a), it can be seen

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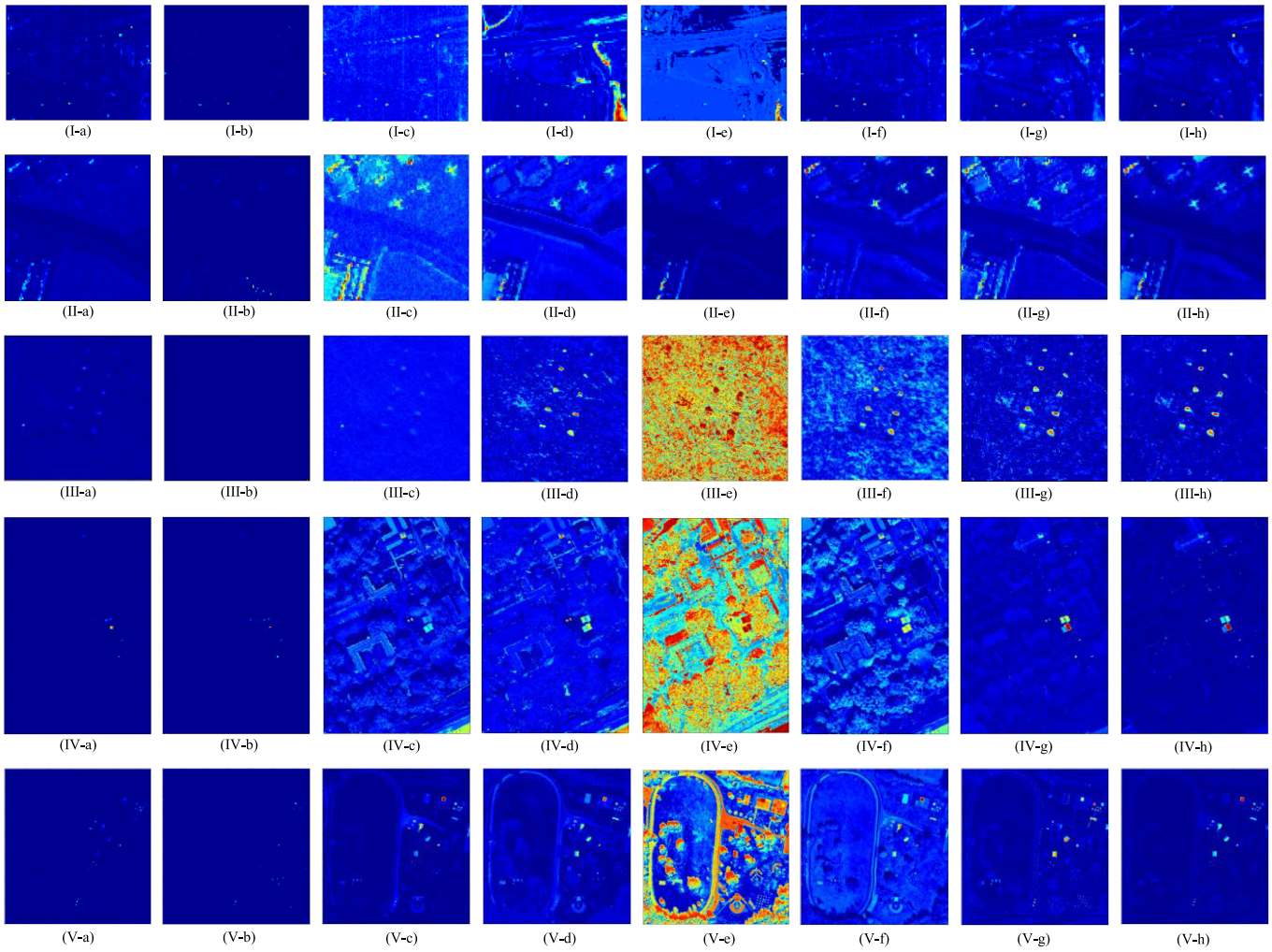


Fig. 6. (I) Detection maps of different methods on the HYDICE Urban data set. (II) Detection maps of different methods on the AVIRIS San Diego data set. (III) Detection maps of different methods on the Cri hyperspectral data set. (IV) Detection maps of different methods on the MUUFL Gulfport data set. (V) Detection maps of different methods on the Mosaic Avon data set. (a) GRX. (b) LRX. (c) LSDM-MoG. (d) AE. (e) ADLR. (f) LRASR. (g) LRDL. (h) SDLRDL.

that the detection probabilities of the SDLRDL methods are higher than the other methods when the false alarm rate is smaller than 10^{-3}. The detection probabilities of the LRDL and SDLRDL methods are approximately 1 when the false alarm rate is about 10^{-1}. It illustrates the better performance of dictionary-learning-based methods on the data set which includes point anomaly. In Fig. 7(b), when the false alarm rate is equal to 10^{-2}, the detection probability of the LRASR method is higher than 0.9. In contrast, the detection probabilities of our proposed methods are higher than 0.95 when the false alarm rate is equal to 10^{-1}, which demonstrates the comparable performance of the methods based on low-rank and sparse representation in the data set with complex background. In Fig. 7(c) and (d), the detection probabilities of LRDL and SDLRDL are obviously higher than the other methods, which shows the distinct advantage of the dictionary-learning-based methods on the data set that contains anomaly with big size. We can observe from Fig. 7(e) that the GRX and our proposed methods can achieve acceptable performance which are better than the other methods, which is beneficial for the estimation of background from global.

The experimental results demonstrate that our proposed SDLRDL method can achieve credible performances on different data sets. First, the introduced spectral-difference operator considered the redundancy of the adjacent bands and it can effectively remove the similarity noise between the adjacent bands to learn a more reliable dictionary. Then, the learned dictionary with spectral-difference low-rank constraint can well-exploit the low-rank prior for the reconstruction of background. Moreover, the $\ell_{2,1}$ norm on the residual matrix $\mathbf{E}$ can positively explore the structural sparsity to detect anomaly.

C. Effectiveness Analysis

To evaluate the effectiveness of the proposed method, the intrinsic characteristic of the proposed method are explored on the HYDICE Urban data set in Fig. 8. We select several spectra of background pixels to display in Fig. 8(I-a). The spatial horizontal coordinates and vertical coordinates of these pixels are {1, 20, 40, 60, 80, 100} and {1, 20, 40, 60, 80}, respectively, and the pixel with spatial coordinates (1, 80) is removed. The spectral curves of anomalous pixels are

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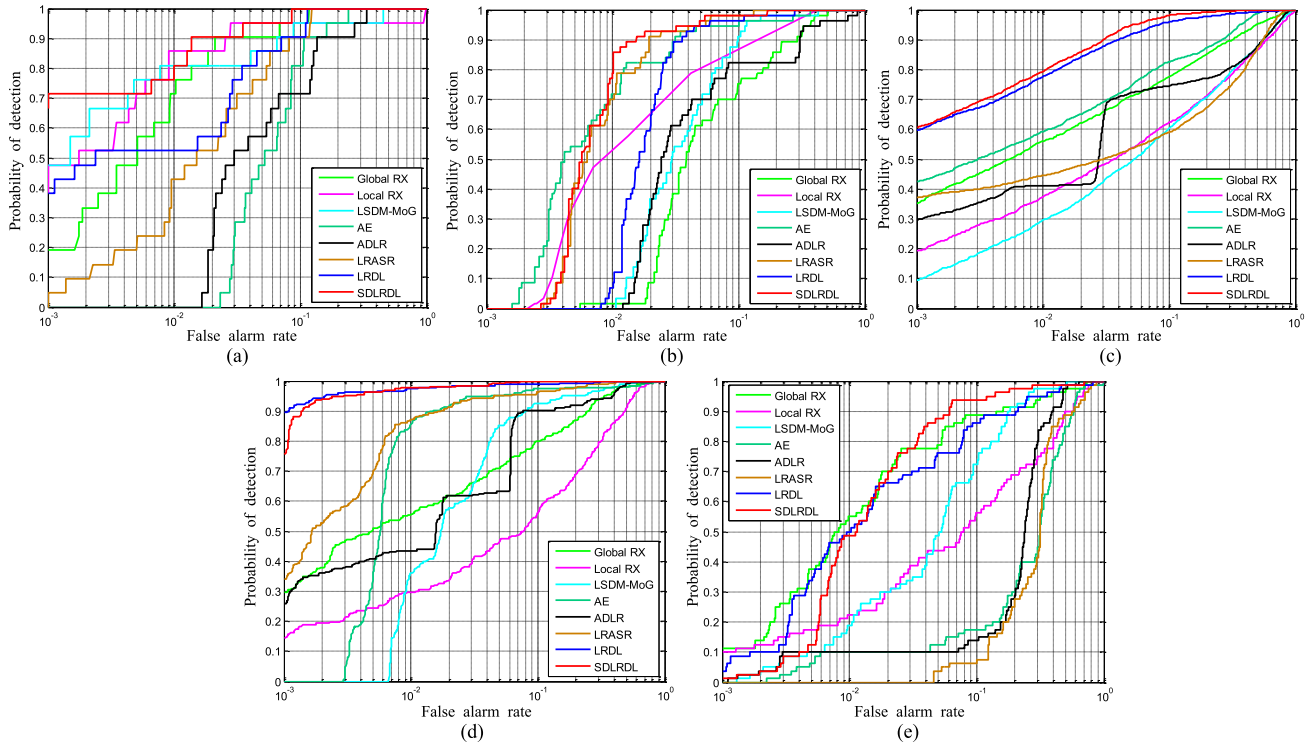


Fig. 7. ROC curves of different methods on the five data sets. (a) HYDICE Urban data set. (b) AVIRIS San Diego data set. (c) Cri data set. (d) MUUFL Gulfport data set. (e) Mosaic Avon data set.

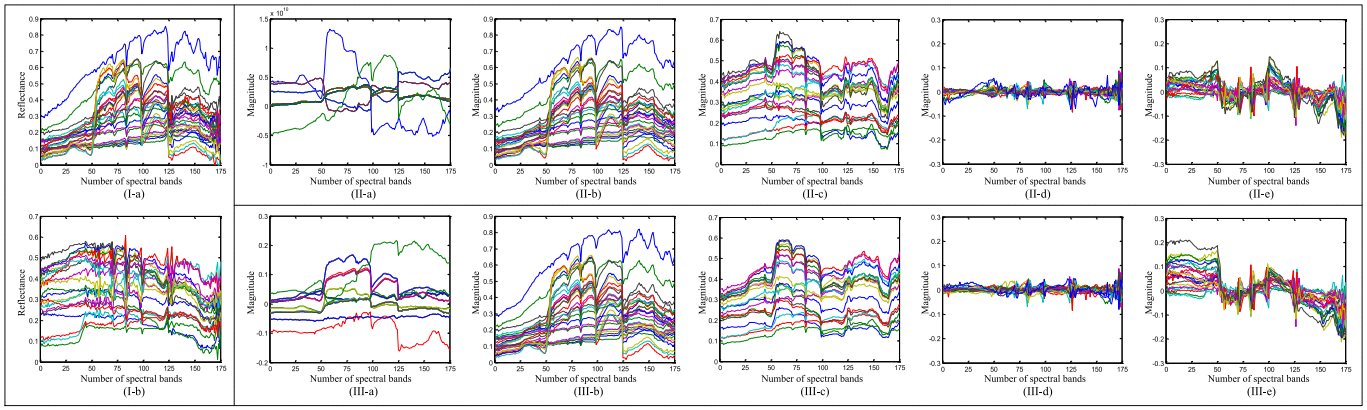


Fig. 8. HYDICE Urban data set. (I-a) and (I-b) Spectral curves about the selected background pixels and anomalous pixels, respectively. (II) and (III) LRDL and SDLRDL methods, in which (a)–(c) are respectively the curves of the learned dictionary, the spectral curves of the reconstructed background pixels, and the spectral curves of the reconstructed anomalous pixels. (d) and (e) Error response about background pixels and anomalous pixels, respectively.

described in Fig. 8(I-b), and we can observe that the spectral ranges of background and anomaly are overlapped, which makes some pixels be falsely detected or missed. The curves of the learned dictionary of the LRDL and SDLRDL methods are respectively shown in Fig. 8(II-a) and (III-a), in which the latter one is more stable. Fig. 8(II-b) and (III-b) are the spectral curves of the reconstructed background pixels by our proposed methods. The background pixels can be well-reconstructed, which demonstrates the positive effect of our proposed methods. But the anomalous pixels in Fig. 8(II-c) and (III-c) cannot be well-reconstructed in comparison to the real spectral curves of anomalous pixels in Fig. 8(I-b). The learned dictionary can better describe the background rather than the anomaly, and the error responses in reconstructing the background are with low magnitude which can be observed in Fig. 8(II-d) and (III-d). Contrarily, the responses of the anomaly in Fig. 8(II-e) and (III-e) are salient. The responses in Fig. 8(d) and (e) are obviously different, which is beneficial to discriminate the background and anomaly.

In addition, although both LRDL and SDLRDL can well-reconstruct the background, the error responses in Fig. 8(III-d) are more compact especially on the first 50 bands compared with the curves in Fig. 8(II-d), and the responses of the anomalous pixels in Fig. 8(II-e) are much more discriminative

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TABLE II
AUC VALUES OBTAINED VIA ℓ_1 NROM AND $\ell_{2,1}$ NROM OF E OF THE SDLRDL METHOD ON DIFFERENT DATA SETS

	HYDICE Urban	AVIRIS San Diego	Cri	MUUFL Gulfport	Mosaic Avon
ℓ_1 norm	0.9785	0.9772	0.9535	0.9676	0.8993
$\ell_{2,1}$ norm	0.9921	0.9885	0.9887	0.9985	0.9670

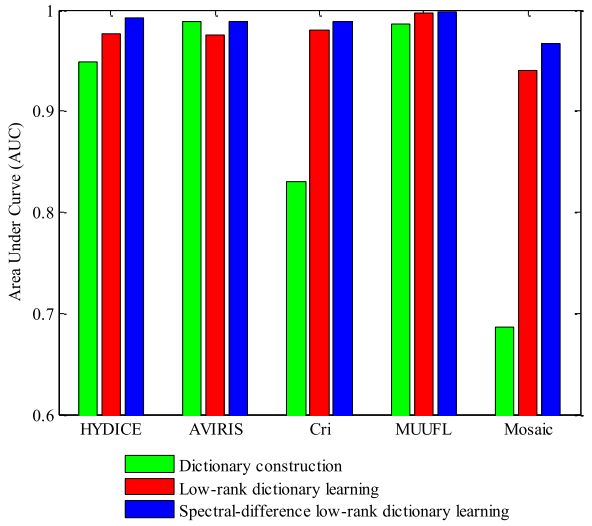


Fig. 9. AUC values on each data set. The results of green bars are obtained by the dictionary construction-based method LRASR. The results of red bars are obtained by the LRDL model of (4), and the results of blue bars are obtained by the SDLRDL model of (6).

than the curves in Fig. 8(II-e). It suggests that the proposed SDLRDL method can provide a more favorable impact on discriminating the background and anomaly than the LRDL method. This is because the SDLRDL method can learn a more stable and robust dictionary to accurately describe the background for hyperspectral anomaly detection.

In our proposed SDLRDL model of (4), the $\ell_{2,1}$ norm is applied on $\mathbf{E}$ to capture the structural sparsity to explore the sample-specific anomalous pixel. In HSI anomaly detection based on the representation of matrix, the $\ell_{2,1}$ norm is much more applicable than the ℓ_1 norm, and the AUC values via the $\ell_{2,1}$ norm and ℓ_1 norm on different data sets are presented in Table II. The experimental results show that the ℓ_1 norm is effective for small data sets, such as the first two data sets. But for the large data sets of the last three data sets, the performance of the $\ell_{2,1}$ norm is well above the ℓ_1 norm. For a large data set with too many pixels, the $\ell_{2,1}$ norm is much more expert in exploring the sample-specific anomalous pixels. Considering the positive characteristic in exploiting the structural sparsity, the $\ell_{2,1}$ norm is applied in our model.

Moreover, the performances of the LRASR method based on dictionary construction and the proposed dictionary-learning-based methods including LRDL and SDLRDL are compared in Fig. 9. We can observe that the proposed dictionary-learning-based methods can play a positive role on different data sets, and this is because the background features can be well-extracted via dictionary learning for background reconstruction to discriminate anomaly. As the dictionary construction method cannot ensure the dictionary includes comprehensive background spectra, and the detection results rely on the constructed dictionary, the proposed dictionary-learning-based methods are more effective. From the red and blue bars, we can observe that the spectral-difference low-rank constraint on dictionary can better improve the detection results, and this is because the applied spectral-difference low-rank constraint can make the learned dictionary more stable and the response of anomaly much more useful for the detection.

D. Computation Costs and Parameters Analysis

The experiments are performed on a computer with 64-b 8-core Intel i7-7700 CPU 3.60 GHz and 16.0 GB RAM under Windows 10. The test software we adopted is MATLAB R2014a. We repeat each group of experiments ten times. The average computational costs of different methods on the five data sets and the settings of parameters are presented in Table III. The computational model of GRX is simple, and it has higher execution efficiency than the other methods. The LRX method adopts the dual window to estimate the local background of the pixel under test, and the computational costs are higher than GRX. The decomposition or representation-based methods need more computational time than GRX. The LRASR method consumes more time in the dictionary construction and the iterative solution with the updating and strict ending conditions. For the proposed LRDL and SDLRDL methods, the computational costs are acceptable, especially for the LRDL which is faster than LSDM-MoG, ADLR, and LRASR methods. The LRDL method is more efficient than SDLRDL as the method of SDLRDL would need more time in solving the spectral-difference low-rank.

The parameters' settings of the comparison algorithms on different data sets are presented in Table III. Considering the difference between the data sets, the dual windows (W_{in}, W_{out}) of LRX are set to different values to obtain the optimal result. In LSDM-MoG, the initial value of mixture Gaussian noise K is set to be 4 for the five data sets as described in [14]. Vary the initial rank l_0 of LSDM-MoG from 10 to 100 with an interval 10 and the optimal value is denoted. The dimension of the latent representation $n_{\mathbf{z}}$ is an important parameter in the AE-based method and it is tuned in [3, 5, 8, 10, 15]. In the experiment of the ADLR method, c is the parameter of endmembers for unmixing and tuned in [10, 20], the bandwidth bw in clustering for dictionary construction is evaluated in [0.05, 1], the parameter λ changes in [0.01, 1] to balance the low-rank and sparse part in low-rank and sparse decomposition, and the optimal results of ADLR are recorded. In the dictionary construction of the LRASR method, just as the settings in [30], the numbers of clusters and selected atoms from each cluster are set to $K = 15$ and $J = 20$, respectively; the weighted parameter on the ℓ_1 norm denoted as β and the parameter on the $\ell_{2,1}$ norm denoted as λ are both set to 0.1 on different data sets except the Cri hyperspectral data set. The performance of LRASR is sensitive to β and not sensitive to

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TABLE III
PARAMETERS SETTINGS AND RUNNING TIME COMPARISONS

		HYDICE Urban	AVIRIS San Diego	Cri	MUUFL Gulfport	Mosaic Avon
GRX	Times	0.459	0.480	1.159	0.793	4.605
LRX	Parameters	$W_{in}=7$ $W_{out}=9$	$W_{in}=11$ $W_{out}=29$	$W_{in}=17$ $W_{out}=19$	$W_{in}=13$ $W_{out}=17$	$W_{in}=11$ $W_{out}=15$
	Times	27.451	66.135	80.387	52.520	1309.814
LSDM-MoG	Parameters	$l_0=70$ $K=4$	$l_0=10$ $K=4$	$l_0=20$ $K=4$	$l_0=10$ $K=4$	$l_0=10$ $K=4$
	Times	209.948	17.202	81.428	39.347	283.228
AE	Parameters	$n_z=5$	$n_z=5$	$n_z=3$	$n_z=3$	$n_z=10$
	Times	126.775	177.826	71.955	67.811	5657.072
ADLR	Parameters	$c=15$ $bw=0.5$ $\lambda=0.01$	$c=10$ $bw=1$ $\lambda=0.01$	$c=15$ $bw=0.5$ $\lambda=0.1$	$c=15$ $bw=1$ $\lambda=0.01$	$c=10$ $bw=1$ $\lambda=0.01$
	Times	15.488	12.863	408.777	155.630	202.746
LRASR	Parameters	$\beta=0.1$ $\lambda=0.1$	$\beta=0.1$ $\lambda=0.1$	$\beta=0.5$ $\lambda=0.1$	$\beta=0.1$ $\lambda=0.01$	$\beta=0.1$ $\lambda=0.1$
	Times	241.713	276.104	732.872	446.635	1013.474
LRDL	Parameters	$m=30$ $\alpha=1$ $\beta=0.1$	$m=30$ $\alpha=0.01$ $\beta=0.01$	$m=30$ $\alpha=0.1$ $\beta=1$	$m=30$ $\alpha=1$ $\beta=1$	$m=30$ $\alpha=1$ $\beta=1$
	Times	16.184	14.090	191.675	115.522	369.478
SDLRDL	Parameters	$m=30$ $\gamma=1$ $\tau=20$	$m=50$ $\gamma=10$ $\tau=10$	$m=50$ $\gamma=1$ $\tau=25$	$m=30$ $\gamma=1$ $\tau=20$	$m=30$ $\gamma=1$ $\tau=10$
	Times	21.662	30.801	339.283	127.609	440.724

λ, and thus we set $\beta = 0.5$ and $\lambda = 0.1$ to obtain optimal result for the Cri hyperspectral data set.

Moreover, the influences of the regularization parameters in the proposed LRDL and SDLRDL methods on the performances of different data sets are analyzed in Fig. 10. In the LRDL method, the parameters α and β are used to adjust the weights of the low-rank term and the $\ell_{2,1}$ norm term, respectively. In the SDLRDL method, the parameters γ and τ are used to control the weights of the spectral-difference low-rank part and the $\ell_{2,1}$ norm part, respectively. The changes in the AUC values with jointly tuning the parameters of α and β on different data sets can be observed from the top row of Fig. 10, in which α and β are both set as {0.001, 0.01, 0.1, 1, 5, 10}, respectively. For the HYDICE Urban data set of Fig. 10(I-a), the obtained AUC values are stable with α varying from 0.001 to 10 when β is more than 0.1. From Fig. 10(II-a), we can observe that the performance of the LRDL method on the San Diego data sets is sensitive to the parameters α and β, and the obtained results are acceptable when $\alpha = 0.01$ and $\beta \in [0.001, 1]$. The changes in the AUC values in Fig. 10(III-a) do not reflect more salient characteristic on the Cri hyperspectral data set. For the MUUFL Gulfport data set of Fig. 10(IV-a), the results are enjoyable and all more than 0.9. In Fig. 10(V-a), the Mosaic Avon data set is sensitive to β and the results are acceptable when $\beta \in [0.1, 1]$.

The figures of the second row in Fig. 10 describe the performances of the SDLRDL method by setting the parameters of γ and τ with different values on the five data sets. Except for Fig. 10(III-b), γ and τ of the other figures in the second row of Fig. 10 are both set as {0.1, 1, 10, 20, 30, 40}, respectively. In Fig. 10(III-b), γ and τ are both set as {0.1, 1, 10, 15, 20, 25}, respectively. We can see from Fig. 10(I-b) that the SDLRDL method is feasible on the HYDICE Urban data set when $\gamma \in [0.1, 20]$ and $\tau \in [10, 40]$. For the San Diego data set, the SDLRDL method can obtain satisfactory results when $\gamma \in [10, 40]$ and $\tau \in [1, 40]$. In Fig. 10(III-b), the performance of the SDLRDL method is stable, and especially, the obtained results are pleasurable when $\tau \in [20, 25]$. For MUUFL Gulfport data set of Fig. 10(IV-b), the SDLRDL method is not sensitive to the parameters and the obtained AUC values are desirable which are all more than 0.95. When $\tau \in [10, 40]$ in Fig. 10(V-b), the AUC values of the Mosaic Avon data set are stable and pleasurable which are all more than 0.90.

In addition, the analyses about the number of the dictionary atoms m of LRDL and SDLRDL methods are presented in Fig. 11, in which m is set as {5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100}. For the LRDL method, the obtained AUC values are more than 0.95 on the HYDICE Urban data set in Fig. 11(I-a) with m ranging from 5 to 100, which demonstrates the pleasurable performance on the data set which includes point anomaly and complicated background. Fig. 11(II-a) reflects the influence of m on the San Diego data set, in which the performance of the LRDL method is acceptable when m ranges from 10 to 60 and the obtained AUC values are more than 0.9. In Fig. 11(III-a), the obtained AUC values on the Cri hyperspectral data set increase when m ranges from 5 to 30; while the results decrease with the increase in the dictionary atoms when m is greater than 30. This is because the background component is not so complex and can be represented with a dictionary which includes limited background spectral features. The results in Fig. 11(IV-a) reflect the stable performance on the MUUFL Gulfport data set. As shown in Fig. 11(V-a), the results are acceptable when m ranges in [5, 40]. The SDLRDL method obtains enjoyable performance on the HYDICE Urban data set when m ranges in [10, 60] in Fig. 11(I-b). It can be seen from Fig. 11(II-b) that the obtained AUC values on San Diego data set are more than 0.95 with m ranging in [5, 50]. The result of the SDLRDL method is acceptable on the Cri hyperspectral data set with m varying from 10 to 100. Fig. 11(IV-b) and (V-b) depicts

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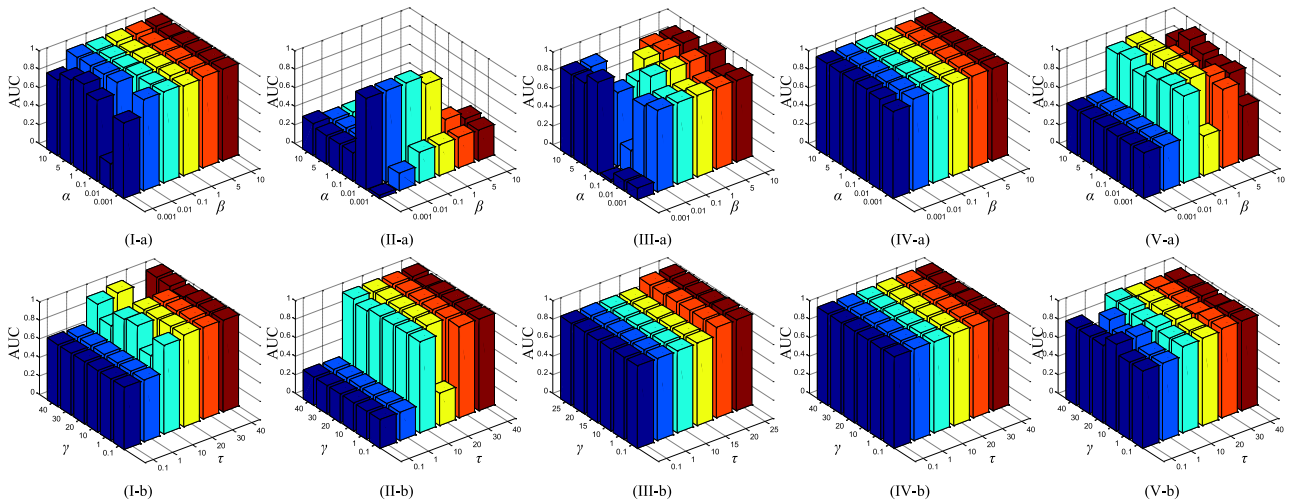


Fig. 10. Analysis of parameters α and β of the LRDL method and γ and τ of the SDLRDL method on each data set. (I-a) and (I-b) HYDICE Urban data set with the LRDL and SDLRDL methods, respectively. (II-a) and (II-b) AVIRIS San Diego data set with the LRDL and SDLRDL methods, respectively. (III-a) and (III-b) Cri hyperspectral data set with the LRDL and SDLRDL methods, respectively. (IV-a) and (IV-b) MUUFL Gulfport data set with the LRDL and SDLRDL methods, respectively. (V-a) and (V-b) Mosaic Avon data set with the LRDL and SDLRDL methods, respectively.

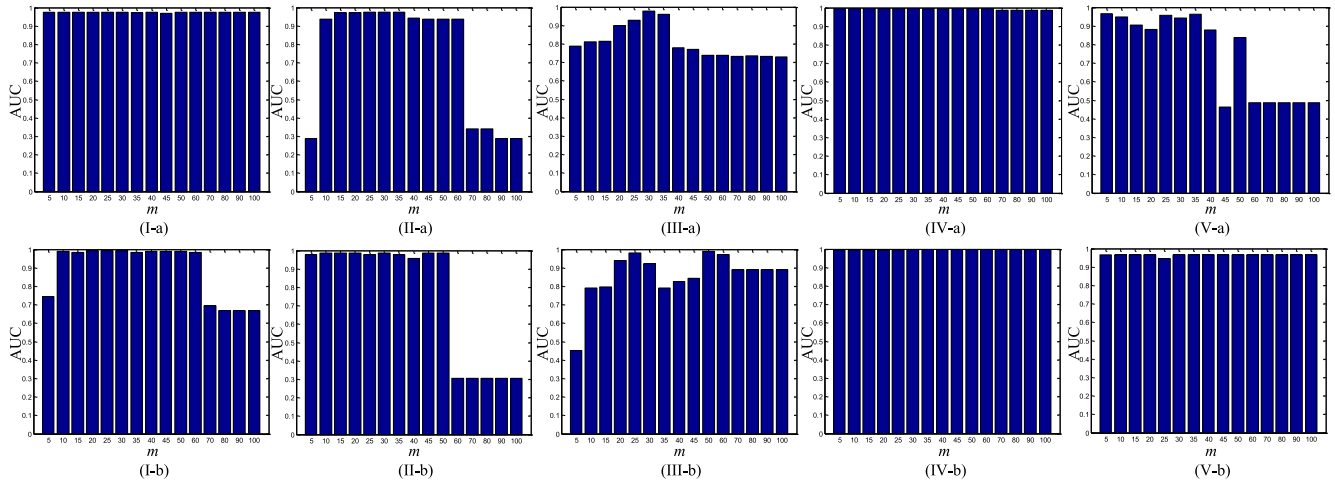


Fig. 11. Analysis of the number of dictionary atoms m of the LRDL method and SDLRDL method on each data set. (I-a) and (I-b) HYDICE Urban data set with the LRDL and SDLRDL methods, respectively. (II-a) and (II-b) AVIRIS San Diego data set with the LRDL and SDLRDL methods, respectively. (III-a) and (III-b) Cri hyperspectral data set with the LRDL and SDLRDL methods, respectively. (IV-a) and (IV-b) MUUFL Gulfport data set with the LRDL and SDLRDL methods, respectively. (V-a) and (V-b) Mosaic Avon data set with the LRDL and SDLRDL methods, respectively.

the pleasurable performance of the SDLRDL method on the MUUFL Gulfport and Mosaic Avon data sets, respectively, as the limited number of dictionary atoms can well-reconstruct the background, and the AUC values are all more than 0.90.

V. Conclusion

In this article, the low-rank representation learning methods are proposed for hyperspectral anomaly detection. In our model, the HSI data are decomposed into background and anomaly, where the background is represented by the product of a dictionary and coefficients matrix via iterative learning, and anomaly is included in the residual part. In the formulation, the low-rank and sparse prior are imposed. Compared with the state-of-the-art methods based on dictionary construction, our proposed methods do not have to build dictionary in a separated step. Moreover, as the spectral redundancy between the bands in HSI data, the spectral-difference low-rank representation learning method was also described. The experimental results on five real data sets demonstrate the effectiveness of our proposed dictionary-learning-based methods. Considering the superior advantages of deep learning in feature extraction, we will explore the deep feature learning for hyperspectral anomaly detection in our future work.

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